

A **random variable** is a function that associates a real number with each element in the sample space.

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values y of the random variable Y , where Y is the number of red balls, are

Sample Space	y
RR	2
RB	1
BR	1
BB	0



Random variable and its types

Random variable:

It is a rule or function that assigns a real no. to each outcome of a sample space.

Types

Discrete
Random Variable

Continuous
Random Variable

* It can take only finite or countable no. of values.

* Ex. 1. Daily count of students in a class

2. No. of phone calls received per day

3. No. of passengers travelling through metro per day.

* It can all possible values between certain limits.

* Ex. 1. Temperature at various points.

2. Age

3. Height

4. Weight

Probability Mass Function :

$$P[X = x_i] = p(x_i) \quad \text{s.t.}$$

$$(i) \quad p(x_i) \geq 0$$

$$(ii) \quad \sum p(x_i) = 1$$

Probability distribution:

X	x_1	x_2	x_3	$\dots$
$p(x)$	$p(x_1)$	$p(x_2)$	$p(x_3)$	$\dots$

A balanced coin is tossed twice. Let X denote the number of heads. Find the probability mass function of X .

$$S = \{ \text{HH}, \text{HT}, \text{TH}, \text{TT} \}$$

$$X = \{ 0, 1, 2 \}$$

$\frac{\text{favourable outcome}}{\text{total outcome}}$

$$P(X=0) = p(0) = \frac{1}{4}$$

$$P(X=1) = p(1) = \frac{2}{4} = \frac{1}{2}$$

$$P(X=2) = p(2) = \frac{1}{4}$$

Ex 3: Check whether the following are probability distributions or not:

(i)	x	0	1	✓
	$p(x)$	$\frac{1}{2}$	$\frac{3}{4}$	✓

Sol. $p(x_i) \geq 0$

$$\sum p(x_i) = \frac{1}{2} + \frac{3}{4} = \frac{5}{4} > 1$$

This is not a probability distribution.

Ex 4.: For the following probability distribution of a discrete random variable X , find
(i) the constant c , (ii) $P(X \leq 3)$ (iii) $P(1 < X < 4)$:

X	0	1	2	3	4	5
$p(x)$	0	c	c	$2c$	$3c$	c

Sol. (i) $\sum p(x_i) = 1$

$$0 + c + c + 2c + 3c + c = 1$$

$$8c = 1$$

$$c = \frac{1}{8}$$

$$\begin{aligned}
 \text{(ii)} \quad P(X \leq 3) &= p(0) + p(1) + p(2) + p(3) \\
 &= 0 + \frac{1}{8} + \frac{1}{8} + \frac{2 \times 1}{8} \\
 &= \frac{4}{8}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad P(1 < X < 4) &= p(2) + p(3) \\
 &= \frac{1}{8} + \frac{2}{8} \\
 &= \frac{3}{8}
 \end{aligned}$$

3.5 Determine the value c so that each of the following functions can serve as a probability distribution of the discrete random variable X :

(a) $f(x) = c(x^2 + 4)$, for $x = 0, 1, 2, 3$;

(b) $f(x) = c \binom{2}{x} \binom{3}{3-x}$, for $x = 0, 1, 2$.

To determine the value of c so that each function serves as a probability distribution, the sum of $f(x)$ over all possible values of x must equal 1:

$$\sum_x f(x) = 1.$$

(a) $f(x) = c(x^2 + 4)$, for $x = 0, 1, 2, 3$

1. Sum up $f(x)$ for all values of x :

$$\sum_{x=0}^3 f(x) = c \sum_{x=0}^3 (x^2 + 4).$$

2. Calculate the summation:

$$\sum_{x=0}^3 (x^2 + 4) = \sum_{x=0}^3 x^2 + \sum_{x=0}^3 4.$$

- $\sum_{x=0}^3 x^2 = 0^2 + 1^2 + 2^2 + 3^2 = 0 + 1 + 4 + 9 = 14$.
- $\sum_{x=0}^3 4 = 4 + 4 + 4 + 4 = 16$.

$$\sum_{x=0}^3 (x^2 + 4) = 14 + 16 = 30.$$

$$c \cdot 30 = 1 \quad \implies \quad c = \frac{1}{30}.$$

(b) $f(x) = c \cdot 2^x \cdot 3^{3-x}$, for $x = 0, 1, 2$

1. Sum up $f(x)$ for all values of x :

$$\sum_{x=0}^2 f(x) = c \sum_{x=0}^2 (2^x \cdot 3^{3-x}).$$

2. Calculate the summation:

$$\sum_{x=0}^2 (2^x \cdot 3^{3-x}) = (2^0 \cdot 3^3) + (2^1 \cdot 3^2) + (2^2 \cdot 3^1).$$

- $2^0 \cdot 3^3 = 27,$
- $2^1 \cdot 3^2 = 18,$
- $2^2 \cdot 3^1 = 12.$

3. Set the total probability to 1:

$$c \cdot 30 = 1 \quad \implies \quad c = \frac{1}{30}.$$

(b) $f(x) = c \cdot 2^x \cdot 3^{3-x}$, for $x = 0, 1, 2$

1. Sum up $f(x)$ for all values of x :

$$\sum_{x=0}^2 f(x) = c \sum_{x=0}^2 (2^x \cdot 3^{3-x}).$$

2. Calculate the summation:

$$\sum_{x=0}^2 (2^x \cdot 3^{3-x}) = (2^0 \cdot 3^3) + (2^1 \cdot 3^2) + (2^2 \cdot 3^1).$$

- $2^0 \cdot 3^3 = 27,$
- $2^1 \cdot 3^2 = 18,$
- $2^2 \cdot 3^1 = 12.$

Probability Density Function



Continuous Random Variable:

A variable that can take all possible values within certain limits.

Probability density function:

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$

where $f(x)$ is the probability function in the interval $[a, b]$.

$f(x)$ is called probability density function

$f(x)$ is called probability density function
iff total probability is unity.

Properties:

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$



Ex. 1:

$f(x) = 6x(1-x)$, $(0 \leq x \leq 1)$
Check whether $f(x)$ is a p.d.f. or not.

Sol.

$$\int_0^1 f(x) dx = \int_0^1 6x(1-x) dx$$

$$= \int_0^1 (6x - 6x^2) dx$$

$$= \left[\frac{6x^2}{2} - \frac{6x^3}{3} \right]_0^1$$

$$= 1 - 0$$

$$= 1$$

$\Rightarrow f(x)$ is a p.d.f.

3.12 An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T , the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0, & t < 1, \\ \frac{1}{4}, & 1 \leq t < 3, \\ \frac{1}{2}, & 3 \leq t < 5, \\ \frac{3}{4}, & 5 \leq t < 7, \\ 1, & t \geq 7, \end{cases}$$

find

- (a) $P(T = 5)$;
- (b) $P(T > 3)$;
- (c) $P(1.4 < T < 6)$;
- (d) $P(T \leq 5 \mid T \geq 2)$.

We will use the properties of the cumulative distribution function (CDF), $F(t)$, to solve the problems.

The CDF, $F(t)$, gives the probability that the random variable T is less than or equal to t , i.e.,

$$F(t) = P(T \leq t).$$

Given CDF:

$$F(t) = \begin{cases} 0, & t < 1, \\ \frac{1}{4}, & 1 \leq t < 3, \\ \frac{1}{2}, & 3 \leq t < 5, \\ \frac{3}{4}, & 5 \leq t < 7, \\ 1, & t \geq 7. \end{cases}$$

(a) $P(T = 5)$

- For a continuous random variable, the probability of T taking an exact value (e.g., $T = 5$) is 0:

$$P(T = 5) = 0.$$

(b) $P(T > 3)$

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- $P(T > 3) = 1 - F(3).$
- From the CDF:

$$F(3) = \frac{1}{2}.$$

- Therefore:

$$P(T > 3) = 1 - \frac{1}{2} = \frac{1}{2}.$$

(c) $P(1.4 < T < 6)$

- $P(1.4 < T < 6) = F(6) - F(1.4).$
- From the CDF:
 - $F(6) = \frac{3}{4}$ (since $5 \leq 6 < 7$),
 - $F(1.4) = \frac{1}{4}$ (since $1 \leq 1.4 < 3$).
- Therefore:

$$P(1.4 < T < 6) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

- Using the conditional probability formula:

$$P(T \leq 5 \mid T \geq 2) = \frac{P(2 \leq T \leq 5)}{P(T \geq 2)}.$$

1. Calculate $P(2 \leq T \leq 5)$:

$$P(2 \leq T \leq 5) = F(5) - F(2).$$

- From the CDF:

- $F(5) = \frac{3}{4},$
- $F(2) = \frac{1}{4}.$

$$P(2 \leq T \leq 5) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

2. Calculate $P(T \geq 2)$:

$$P(T \geq 2) = 1 - F(2).$$

- From the CDF:

- $F(2) = \frac{1}{4}$.

$$P(T \geq 2) = 1 - \frac{1}{4} = \frac{3}{4}.$$

3. Calculate the conditional probability:

$$P(T \leq 5 \mid T \geq 2) = \frac{P(2 \leq T \leq 5)}{P(T \geq 2)} = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}.$$

Ex. 2: A continuous r.v. X has the probability density function $f(x) = Ax^3$, $0 \leq x \leq 1$.

Determine:

(i) A , (ii) $P[0.2 < X < 0.5]$, (iii) $P[X > \frac{3}{4} \text{ given } X > \frac{1}{2}]$.

Sol.:

$$(i) \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^1 f(x) dx = 1$$

$$\Rightarrow \int_0^1 Ax^3 dx = 1$$

$$\Rightarrow A \left[\frac{x^4}{4} \right]_0^1 = 1$$

$$\Rightarrow A \cdot \frac{1}{4} = 1$$

$$\Rightarrow A = 4$$

$$(ii) \quad P[0.2 < x < 0.5] = \int_{0.2}^{0.5} f(x) dx$$

$$= \int_{0.2}^{0.5} 4x^3 dx$$

$$= 4 \left[\frac{x^4}{4} \right]_{0.2}^{0.5}$$

$$= 0.0625 - 0.0016$$

$$= 0.0609$$

$$(ii) \quad P\left[X > \frac{3}{4} \text{ given } X > \frac{1}{2}\right]$$

$$= P\left[X > \frac{3}{4} \mid X > \frac{1}{2}\right]$$

$$= \frac{P\left[X > \frac{3}{4} \cap X > \frac{1}{2}\right]}{P\left[X > \frac{1}{2}\right]}$$

$$= \frac{P\left[X > \frac{3}{4}\right]}{P\left[X > \frac{1}{2}\right]}$$

$$[P(A|B) = \frac{P(A \cap B)}{P(B)}]$$

$$= \frac{\int_{3/4}^1 f(x) dx}{\int_{1/2}^1 f(x) dx} = \frac{I_1}{I_2}$$

$$0 \leq x \leq 1$$

$$I_1 = \int_{3/4}^1 4x^3 dx = 4 \left[\frac{x^4}{4} \right]_{3/4}^1 = 1^4 - \left(\frac{3}{4} \right)^4 = 1 - \frac{81}{256} = \frac{175}{256}$$

A random variable is discrete if it takes on a countable number of distinct values. Examples include:

1. The number of students in a class.
2. The result of rolling a six-sided die (1, 2, 3, 4, 5, 6).
3. The number of heads in 10 coin tosses.

Continuous Random Variables

A random variable is continuous if it can take on any value within a given range, typically uncountably infinite. Examples include:

1. The height of students in a class.
2. The time it takes to complete a task.
3. The temperature in a room.

3.1 Classify the following random variables as discrete or continuous:

X : the number of automobile accidents per year in Virginia.

Y : the length of time to play 18 holes of golf.

M : the amount of milk produced yearly by a particular cow.

N : the number of eggs laid each month by a hen.

P : the number of building permits issued each month in a certain city.

Q : the weight of grain produced per acre.

Discrete Random Variables

These are random variables that take on countable values.

1. X : The number of automobile accidents per year in Virginia.
 - **Reason:** The number of accidents is countable and cannot take fractional values
2. N : The number of eggs laid each month by a hen.
 - **Reason:** The number of eggs is countable and must be a whole number.
3. P : The number of building permits issued each month in a certain city.

Continuous Random Variables

These are random variables that take on uncountably infinite values within a range.

4. **Y:** The length of time to play 18 holes of golf.
 - **Reason:** Time is measured on a continuous scale (e.g., in seconds, minutes, etc.).
5. **M:** The amount of milk produced yearly by a particular cow.
 - **Reason:** Milk is measured on a continuous scale (e.g., in liters or gallons).
6. **Q:** The weight of grain produced per acre.
 - **Reason:** Weight is measured on a continuous scale (e.g., in kilograms or pounds).

If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space**.

If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

- **Discrete:** Like counting apples in a basket (1 apple, 2 apples, 3 apples...).
- **Continuous:** Like measuring water in a glass (it could be 250.1 ml, 250.15 ml, or any amount).

Discrete Probability Distributions

A discrete random variable assumes each of its values with a certain probability. In the case of tossing a coin three times, the variable X , representing the number of heads, assumes the value 2 with probability $3/8$, since 3 of the 8 equally likely sample points result in two heads and one tail. If one assumes equal weights for the simple events in Example 3.2, the probability that no employee gets back the right helmet, that is, the probability that M assumes the value 0, is $1/3$. The possible values m of M and their probabilities are

m	0	1	3
$P(M = m)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

Note that the values of m exhaust all possible cases and hence the probabilities add to 1.

A stockroom clerk returns three safety helmets at random to three steel mill employees who had previously checked them. If Smith, Jones, and Brown, in that order, receive one of the three hats, list the sample points for the possible orders of returning the helmets, and find the value m of the random variable M that represents the number of correct matches.

If S , J , and B stand for Smith's, Jones's, and Brown's helmets, respectively, then the possible arrangements in which the helmets may be returned and the number of correct matches are

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable X if, for each possible outcome x ,

1. $f(x) \geq 0$,
2. $\sum_x f(x) = 1$,
3. $P(X = x) = f(x)$.

Key Characteristics of a Discrete Probability Distribution:

1. Countable Values:

The random variable can take on only a finite or countably infinite number of distinct values. For example, the number of heads in 3 coin tosses or the number of people in a room.

2. Probability Assignment:

Each value of the random variable is associated with a probability. These probabilities must satisfy two conditions:

- The probability for each possible outcome is between 0 and 1, inclusive. (i.e., $0 \leq P(X = x_i) \leq 1$).

- The probability for each possible outcome is between 0 and 1, inclusive. (i.e., $0 \leq P(X = x_i) \leq 1$).
- The sum of all the probabilities for all possible outcomes must be 1. That is, if there are n possible outcomes, then:

$$\sum_{i=1}^n P(X = x_i) = 1$$

3. Probability Mass Function (PMF):

The function that assigns a probability to each possible value of the discrete random variable is called the **probability mass function** (PMF). The PMF is denoted by $P(X = x_i)$, where x_i is one of the possible values that the random variable X can take.

4. Expected Value and Variance:

- **Expected Value (Mean):** The expected value is a measure of the "center" of the distribution and can be computed as:

$$E(X) = \sum_{i=1}^n x_i P(X = x_i)$$

Variance: The variance is a measure of how spread out the values are around the expected value. It is given by:

$$\text{Var}(X) = \sum_{i=1}^n (x_i - E(X))^2 P(X = x_i)$$

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Solution: Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can only take the numbers 0, 1, and

2. Now

$$f(0) = P(X = 0) = \frac{\binom{3}{0} \binom{17}{2}}{\binom{20}{2}} = \frac{68}{95}, \quad f(1) = P(X = 1) = \frac{\binom{3}{1} \binom{17}{1}}{\binom{20}{2}} = \frac{51}{190},$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2} \binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}.$$

Thus, the probability distribution of X is

x	0	1	2
$f(x)$	$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$



If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, find a formula for the probability distribution of the number of cars with side airbags among the next 4 cars sold by the agency.

Key Information:

- Probability of success (car with side airbags), $p = 0.5$
- Probability of failure (car without side airbags), $1 - p = 0.5$
- Number of trials (cars sold), $n = 4$

Binomial Distribution Formula:

The binomial distribution gives the probability of having exactly k successes (cars with side airbags) out of n independent trials (cars sold), where each trial has a success probability p .

The formula for the probability mass function (PMF) of a binomial distribution is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Where:

- $P(X = k)$ is the probability of exactly k successes (cars with side airbags),
- n is the total number of trials (4 cars sold),
- k is the number of successes (cars with side airbags),
- p is the probability of success (0.5),
- $\binom{n}{k}$ is the binomial coefficient, which represents the number of ways to choose k successes from n trials and is calculated as:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Applying the Formula:

For $n = 4$ and $p = 0.5$, the probability distribution for the number of cars with side airbags X among the next 4 cars sold is:

$$P(X = k) = \binom{4}{k} (0.5)^k (0.5)^{4-k} = \binom{4}{k} (0.5)^4$$

This simplifies to:

$$P(X = k) = \binom{4}{k} \times \frac{1}{16}$$

Possible Values of k :

Since X represents the number of cars with side airbags, k can take integer values from 0 to 4 (inclusive). For each value of k , we can calculate the corresponding probability:

1. For $k = 0$ (no cars with side airbags):

$$P(X = 0) = \binom{4}{0} \times \frac{1}{16} = 1 \times \frac{1}{16} = \frac{1}{16}$$

2. For $k = 1$ (1 car with side airbags):

$$P(X = 1) = \binom{4}{1} \times \frac{1}{16} = 4 \times \frac{1}{16} = \frac{4}{16} = \frac{1}{4}$$

3. For $k = 2$ (2 cars with side airbags):

$$P(X = 2) = \binom{4}{2} \times \frac{1}{16} = 6 \times \frac{1}{16} = \frac{6}{16} = \frac{3}{8}$$

4. **For $k = 3$** (3 cars with side airbags):

$$P(X = 3) = \binom{4}{3} \times \frac{1}{16} = 4 \times \frac{1}{16} = \frac{4}{16} = \frac{1}{4}$$

5. **For $k = 4$** (4 cars with side airbags):

$$P(X = 4) = \binom{4}{4} \times \frac{1}{16} = 1 \times \frac{1}{16} = \frac{1}{16}$$

Summary of the Probability Distribution:

The probability distribution for the number of cars with side airbags among the next 4 cars sold is:

- $P(X = 0) = \frac{1}{16}$
- $P(X = 1) = \frac{4}{16} = \frac{1}{4}$
- $P(X = 2) = \frac{6}{16} = \frac{3}{8}$
- $P(X = 3) = \frac{4}{16} = \frac{1}{4}$
- $P(X = 4) = \frac{1}{16}$

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X , defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in R$.
2. $\int_{-\infty}^{\infty} f(x) \, dx = 1$.
3. $P(a < X < b) = \int_a^b f(x) \, dx$.

Example 3.11: Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
(b) Find $P(0 < X \leq 1)$.

Solution: We use Definition 3.6.

- (a) Obviously, $f(x) \geq 0$. To verify condition 2 in Definition 3.6, we have

$$\int_{-\infty}^{\infty} f(x) \, dx = \int_{-1}^2 \frac{x^2}{3} \, dx = \frac{x^3}{9} \Big|_{-1}^2 = \frac{8}{9} + \frac{1}{9} = 1.$$

Chapter 3 Random Variables and Probability Distributions

- (b) Using formula 3 in Definition 3.6, we obtain

$$P(0 < X \leq 1) = \int_0^1 \frac{x^2}{3} \, dx = \frac{x^3}{9} \Big|_0^1 = \frac{1}{9}.$$



